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Evaluation of Factors Determining Flight Delays

ECE-143: Programming for Data Analysis (WI 2026)

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Motivation

- The use of commercial flights has drastically increased in the last 30 years especially in the US, as roughly 900 million passengers flew in 2025.

- Because of the increased demand for flights, several airlines and airports have had to better their infrastructure to handle this influx in air traffic

- 10 years of flight data presenting delays and flight conditions potentially hindering the ability of planes to take off was used

- Analyzing this data set presented a set of key contributing factors to why flights are delayed across the entire US.

Method of Analysis

- Several python modules were used to process the dataset to pull out meaningful data and visualizations (NumPy, Pandas, Matplot)

- The factors within the data set were then processed individually and in combination to determine solo impact and aggregate effects.

- The graphs then helped visualize the strength and importance of each factor in the overall cause of flight delays

Breakdown of Data Set

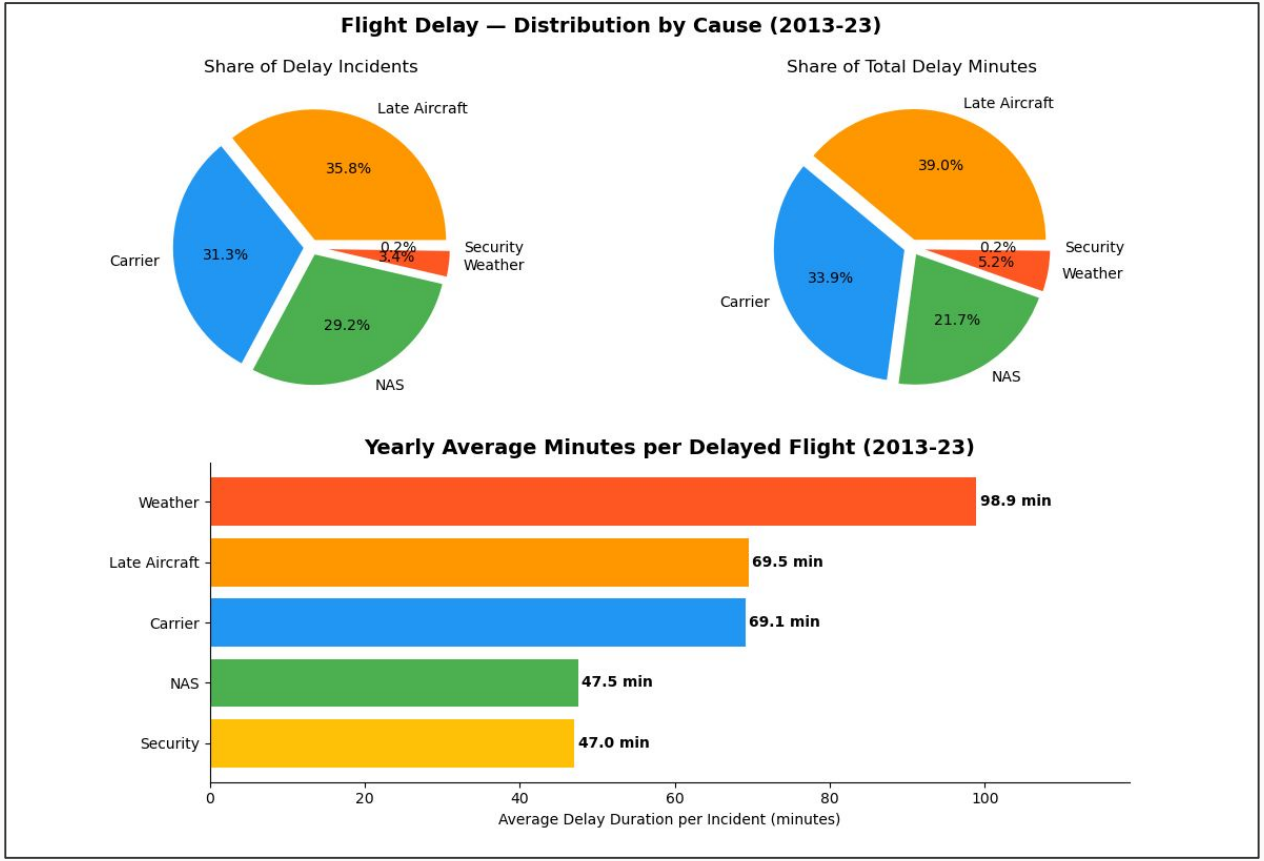
Our data set was retrieved from Kaggle and contained information about the following from the period of 2013 - 2023:

- Month
- Weather delays
- National Airspace System (NAS) delay
- Security delays
- Airline and destination airports
- Diverted, late arrival and cancelled flights
- Number of flights delayed by 15 or more minutes

The data was inside of a large CSV file. This includes all airports with big names like LAX, MCO, and LGA along with airlines like Delta and Southwest.

AIRLINE DELAY ANALYSIS	KEY STATISTICS
Total flight records analysed	171,426
Total arriving flights	62,146,805
Total delayed (≥15 min)	11,375,095
Overall delay rate	18.30%
Total delay minutes	726,758,355
Total cancellations	1,290,923
Overall cancellation rate	2.08%
Unique carriers	21
Unique airports	391
Years covered	2013 - 2023

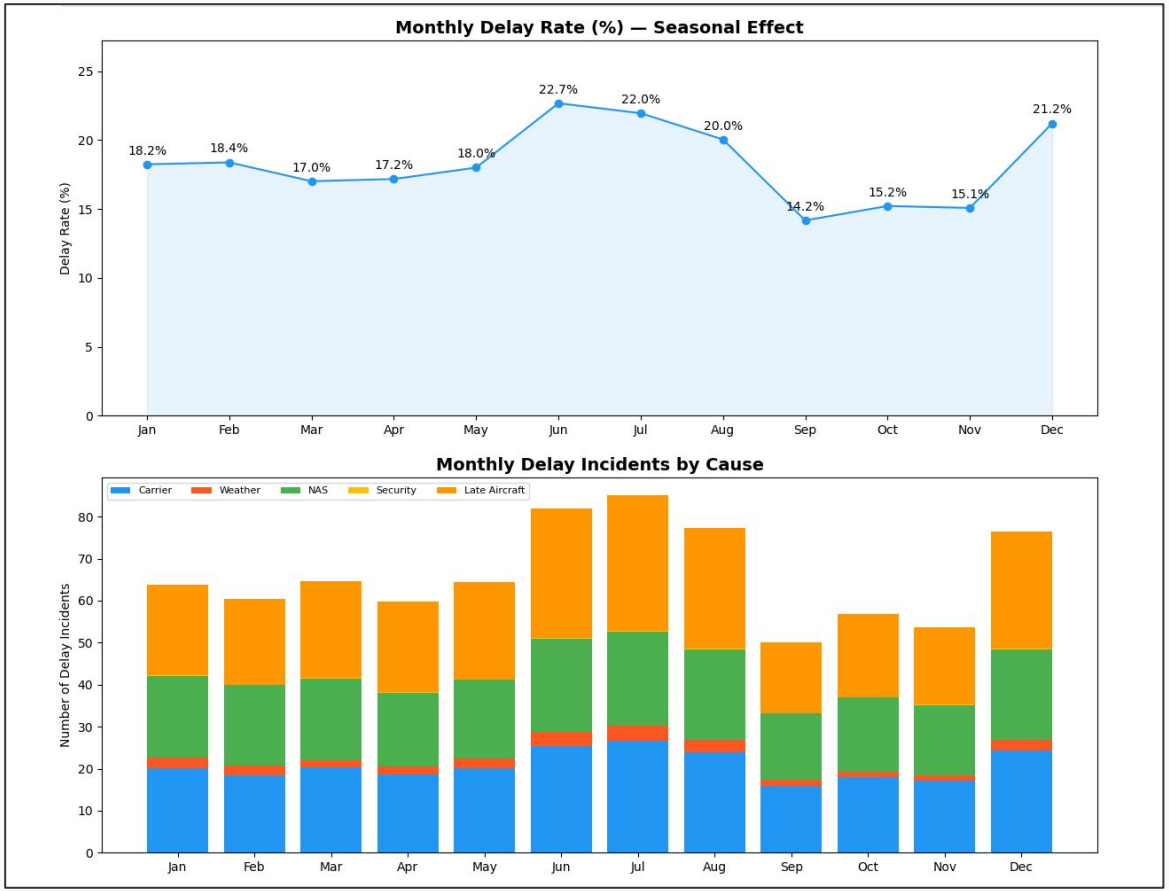
Causes of Delay



- General breakdown of how much the given factors contribute to delays

- Weather is a strong factor of delay yet does not occur enough to compare to the shares which the late arrival, carriers and the NAS hold

Months of the Year

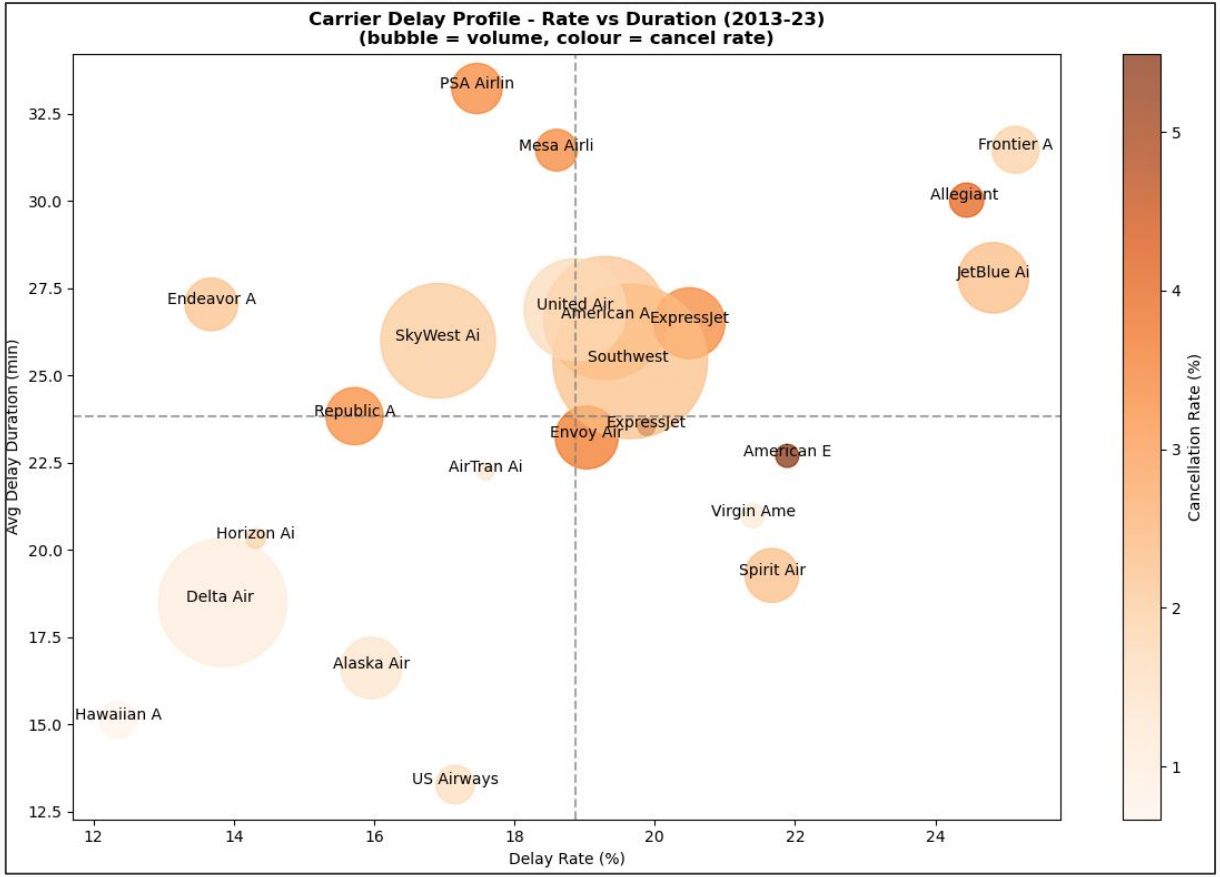


- High delay rates and incidents in the summer months due to increased travelling
- Thicker weather bars in summer attributed to thunderstorms and heat
- December peak caused by holiday traveling and weather storms for most of the US

Year over Year Trends

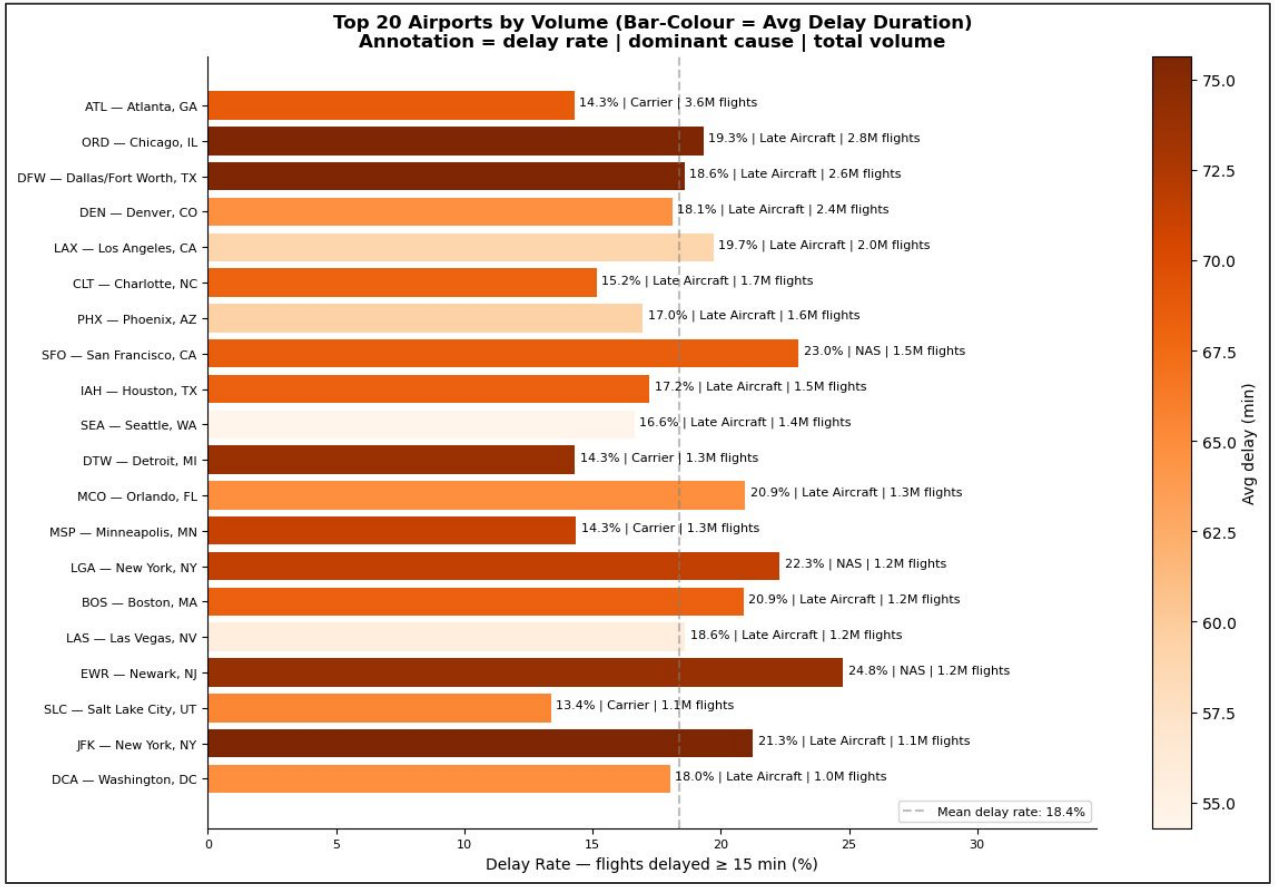
- Delays consistently increasing from 2013 onwards
- Massive flight traffic decrease yields more efficiency, less delay
- Massive delays (200% increase) between 2020-21 showing airlines and NAS were unprepared for burst of travellers

Carrier Abilities



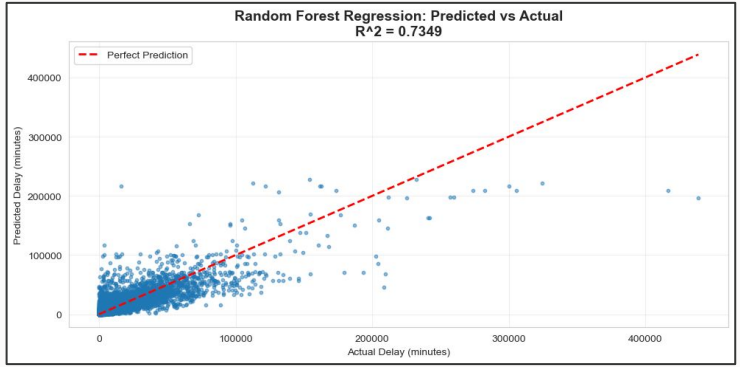
- Demonstration of carrier reliability
- Delta shown to be most reliable major carrier
- Many airlines reside around the averages
- Frontier yields most delayed airline on average

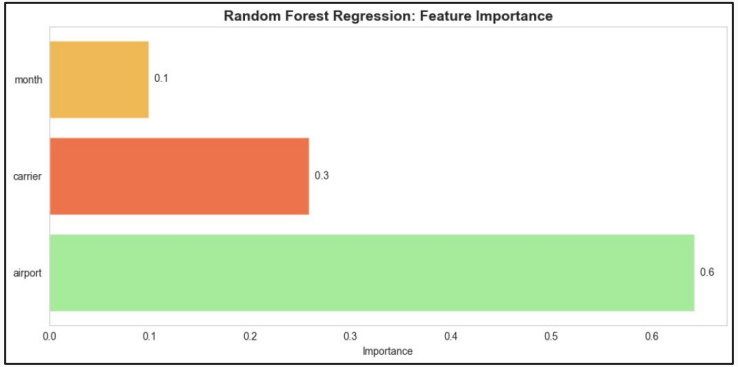
Airport Performance

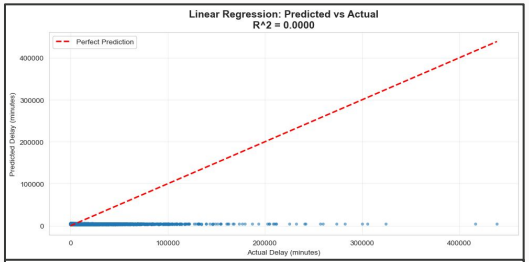


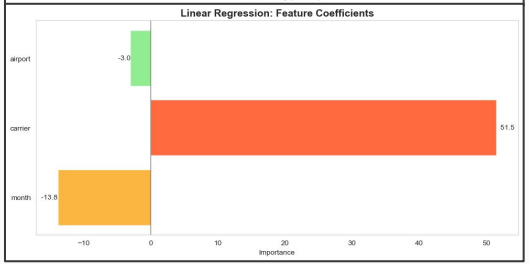
- Top 20 largest airports by volume of flights
- ORD, DFW, EWR, JFK leaders in delay times, >1 hour
- Top performers LAS, PHX, LAX, all high volume airports with near or less than mean delay rate and <1 hour delays

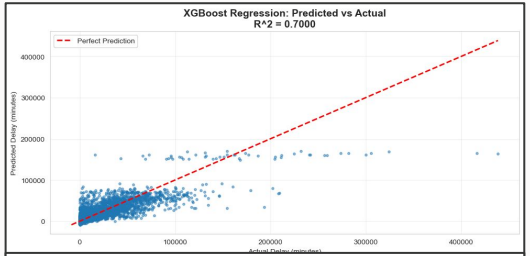
Regression Analysis

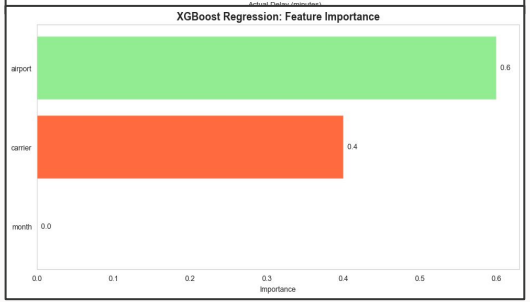












- 3 Regression plots aggregating factors of month, carrier and airport
- Random Forest Regression most insightful
- found airport is most influential delay factor
- Linear and XGBoost regression were less conclusive, yet XGBoost agrees with Random Forest

Future Work

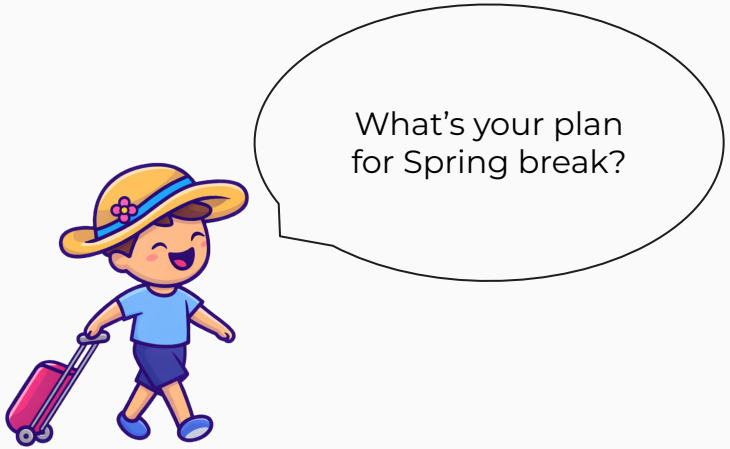
- Future dataset should be aggregated at individual flight and route level in addition to the carrier–airport–month level, so following patterns can be evaluated directly:
  - Route specific analysis.
  - Intraday patterns (such as, early morning vs night flight).
  - Day-of-week effects (weekday vs weekend).

- COVID-19 years (2020–2022) can be evaluated separately using covid specific dataset to evaluate delay and cancellation patterns due to reduced schedules and workforce changes.

- Security delays can be analysed separately as minor security incidents may be absorbed into other categories.

Conclusion

- Late aircraft, carrier operations, and NAS delays are the dominant contributors to flight delays.
- Flight delay shows strong seasonal patterns.
- Carrier performance vary significant.
- Airport is the the most influential factors for flight delay based on our regression results.



Thank you